

## Assignment 1 – *Command Line*

### Description:

The program prints the count of command-line arguments and its corresponding arguments. Count and arguments are subsequently newlined for format.

### Approach:

I know I'll need to print text, so I add `#include <stdio.h>` at the beginning for the function and others. I also know that C has no built-in string type, so I'll have to use an array of pointers that point to each argument, i.e. each char sequence. I knew as well that I would have to format the output correctly, using placeholders and escape sequences.

### Issues and Resolutions:

```
student@student:~/Documents/CSC415-Assignment-1-CommandLine$ make
gcc -c -o Juarez_Julian_HW1_main.o Juarez_Julian_HW1_main.c -g -I.
Juarez_Julian_HW1_main.c: In function 'main':
Juarez_Julian_HW1_main.c:16:9: error: expected declaration or statement at end of input
   16 |         printf("There were %d arguments on the command line.\n", argc);
      |         ^~~~~~
make: *** [Makefile:55: Juarez_Julian_HW1_main.o] Error 1
```

I forgot the closing brace, it resulted in a compile-time error.

```
student@student:~/Documents/CSC415-Assignment-1-CommandLine$ git commit -m "brace added"
[main 5ff8550] brace added
 1 file changed, 1 insertion(+)
student@student:~/Documents/CSC415-Assignment-1-CommandLine$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 286 bytes | 286.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/JulianJua/CSC415-Assignment-1-CommandLine.git
   3499422..5ff8550  main -> main
student@student:~/Documents/CSC415-Assignment-1-CommandLine$ make clean
rm *.o Juarez_Julian_HW1_main
rm: cannot remove '*.o': No such file or directory
rm: cannot remove 'Juarez_Julian_HW1_main': No such file or directory
make: *** [Makefile:61: clean] Error 1
student@student:~/Documents/CSC415-Assignment-1-CommandLine$ make
gcc -c -o Juarez_Julian_HW1_main.o Juarez_Julian_HW1_main.c -g -I.
gcc -o Juarez_Julian_HW1_main Juarez_Julian_HW1_main.o -g -I.
student@student:~/Documents/CSC415-Assignment-1-CommandLine$
```

After adding the brace, compilation was successful.

Analysis: N/A

Screen shot of compilation:

```
student@student:~/Documents/CSC415-Assignment-1-CommandLine$ make  
gcc -c -o Juarez_Julian_HW1_main.o Juarez_Julian_HW1_main.c -g -I.  
gcc -o Juarez_Julian_HW1_main Juarez_Julian_HW1_main.o -g -I.  
student@student:~/Documents/CSC415-Assignment-1-CommandLine$
```

Screen shot(s) of the execution of the program:

```
student@student:~/Documents/CSC415-Assignment-1-CommandLine$ make run RUNOPTIONS="Hello there, how are you?"  
./Juarez_Julian_HW1_main Hello there, how are you?  
  
There were 6 arguments on the command line.  
Argument 00: ./Juarez_Julian_HW1_main  
Argument 01: Hello  
Argument 02: there,  
Argument 03: how  
Argument 04: are  
Argument 05: you?  
student@student:~/Documents/CSC415-Assignment-1-CommandLine$
```